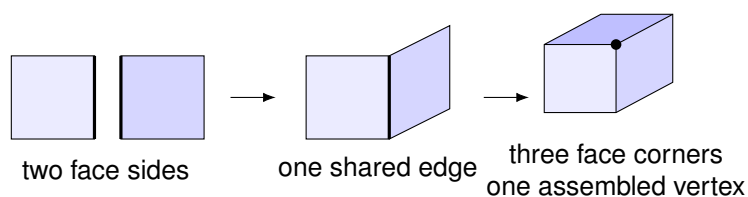
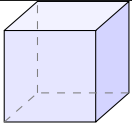
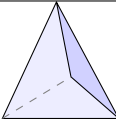
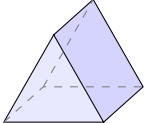
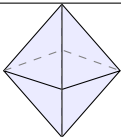


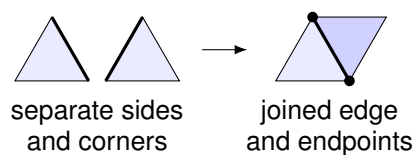
Use closed models with all their faces attached. A vertex is a corner of the assembled solid; an edge is shared by two faces. Count each assembled edge and vertex once. Tabs and glue overlaps do not count.



Problem 1: Find the vertices, edges and faces of each closed model.

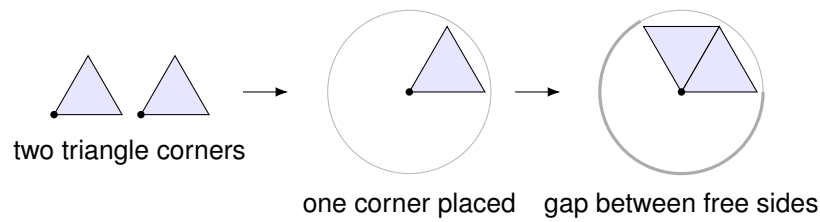
Model	Vertices V	Edges E	Faces F
 cube			
 tetrahedron			
 triangular prism			
 octahedron			

Problem 2: Imagine cutting every face free. Find a rule that turns the number of separate face sides into the assembled edge count. Can the separate face-corner count alone tell you the vertex count?

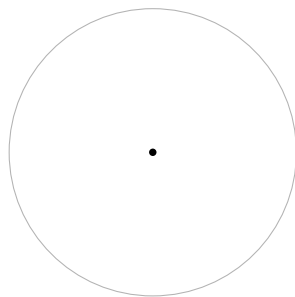


Faces before assembly	Separate sides	Assembled edges	Separate corners	Assembled vertices
6 squares: cube				
4 triangles: tetrahedron				
2 triangles and 3 squares: prism				
8 triangles: octahedron				

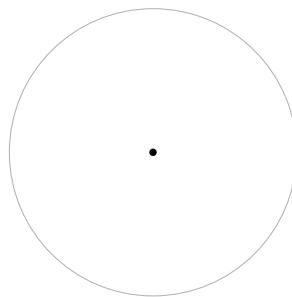
Join face corners around one point. Keep their sides meeting edge to edge and their marked vertices together. The gap is the missing part of a full turn when the corners lie flat.



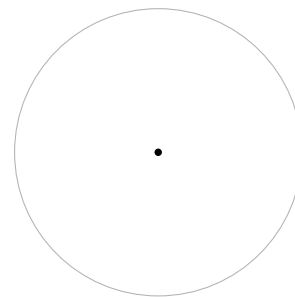
Problem 3: Which groups can close into a pointed corner with no inward dents? Which close flat with no gap? Make each group, then draw its gap with the corners laid flat.



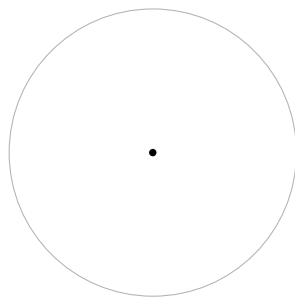
3 triangles



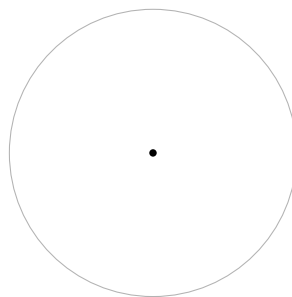
4 triangles



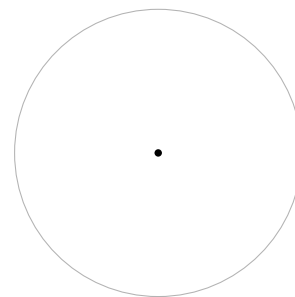
5 triangles



6 triangles

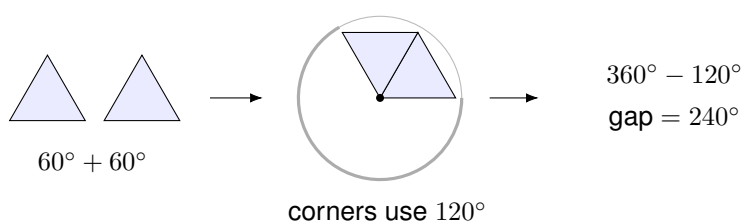


3 squares



4 squares

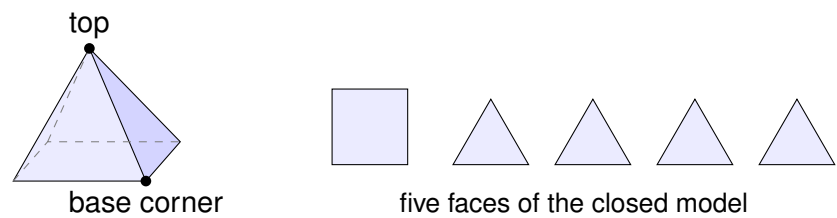
A full turn is 360° . An equilateral triangle corner is 60° ; a square corner is 90° . A vertex's gap is 360° minus its face-corner angles.



Problem 4: Find the gap at every vertex of each closed model. How many full turns do all its gaps make together?

Model	Vertices	Face corners at one vertex	Gap at one vertex	All gaps together
Cube				
Tetrahedron				
Triangular prism				
Octahedron				

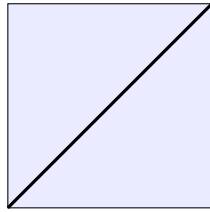
Problem 5: The square pyramid has one square face and four equilateral triangle faces. Find its total gap.



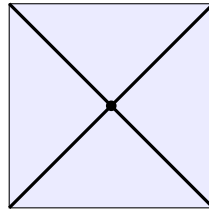
Vertex type	How many?	Face corners meeting there	Gap at each one
Top			
Base corner			

In a redrawn surface, marked points count as vertices, segments between vertices count as edges, and each face piece counts as a face.

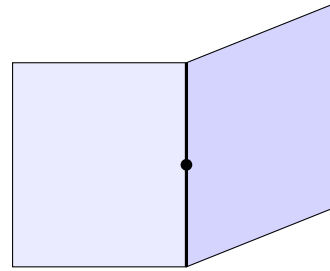
Problem 6: Compare $V - E + F$ for all four models in Problem 1. Keep the cube's shape and try these three changes to its surface drawing. What happens to $V - E + F$ and to the total gap? Can a new vertex have a gap while the cube keeps its shape?



one face divided
by a diagonal



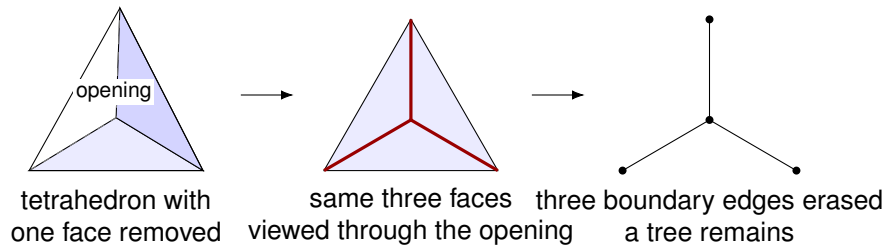
one face divided
from its center



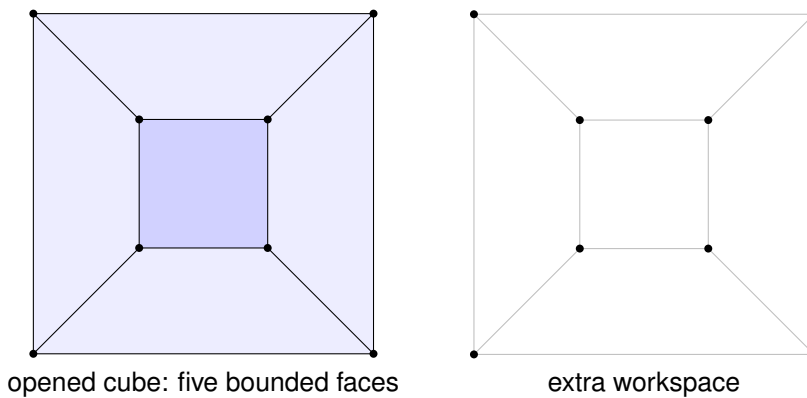
one vertex inserted
along a shared edge

Closed surface	V	E	F	$V - E + F$	Total gap
Original cube					
One diagonal					
Center and 4 lines					
One edge vertex					

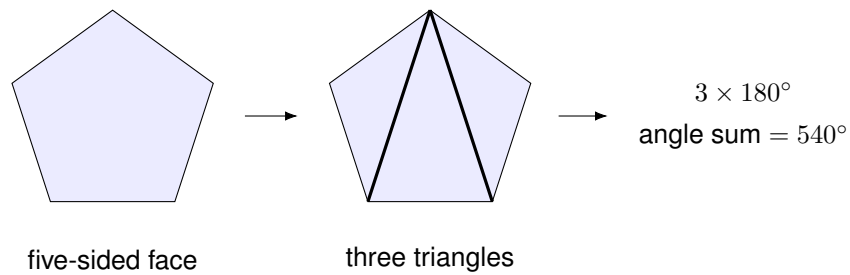
Convex means no dents. An opened drawing preserves the assembled edge and vertex connections. Count its bounded regions as faces; the outside is the removed face. A tree is a connected drawing with no closed loops.



Problem 7: Explain why $V - E + F$ always has the same value for closed convex solids with polygon faces. Can an opened drawing be changed into a tree while keeping all its vertices connected?



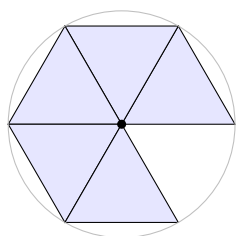
The angles of a triangle add to 180° . A polygon with n sides can be divided into $n - 2$ triangles, so its corner angles add to $180(n - 2)^\circ$.



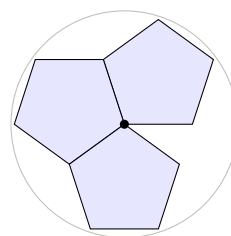
Problem 8: Explain why every closed convex solid with polygon faces has the same total gap, even when its vertex gaps are different.

A regular polygon has equal sides and equal angles. Every face of each solid below is one regular polygon.

Problem 9: Find the number of vertices, edges and faces each closed convex solid with polygon faces must have, without a model of the whole solid.



Triangle faces; five meet
at every vertex



Pentagon faces; three meet
at every vertex

Solid	Vertices	Edges	Faces
Five triangles at each vertex			
Three pentagons at each vertex			